LAB SHEET 4

NAME:RADHIKA

ROLL No:2401010094

Title:

Implementation of User Creation, Deletion, and Privilege Management in SQL.

Problem Statement

Design and execute SQL commands to create users, manage their access privileges, and remove users in a database system. This assignment focuses on database security and access control mechanisms.

Tasks

Part A: User Creation

1. Create a new user named user1 with a password.
2. Create another user user2 with limited access.

```sql
create database LAB_SHEET_4;
USE LAB_SHEET_4;
#part A
CREATE USER 'user1' IDENTIFIED BY 'pass123';
CREATE USER 'user2' IDENTIFIED BY 'pass123';
```

Part B: Granting Privileges

3. Grant **SELECT and INSERT** privileges on a table Student to user1.
4. Grant **ALL privileges** on a table Course to user2.
5. Provide **WITH GRANT OPTION** to user1 and observe its effect.

```
#PART B
GRANT SELECT, INSERT ON Student TO 'user1';
GRANT ALL PRIVILEGES ON Course TO 'user2';
GRANT SELECT, INSERT ON Student TO 'user1' WITH GRANT OPTION;
```

Part C: Revoking Privileges

6. Revoke **INSERT permission** from user1 on the Student table.
7. Revoke all privileges from user2 on the Course table.

```
#C
REVOKE INSERT ON Student FROM 'user1';
REVOKE ALL PRIVILEGES ON Course FROM 'user2';
```

Part D: User Deletion

8. Drop/Delete user user2 from the database.
9. Attempt to login using deleted credentials and observe the result.

```
#D
DROP USER 'user2';
```

Part E: Additional Commands

10. Display all users in the system.
11. Show privileges assigned to a specific user.

```
#E
SELECT user FROM mysql.user;
SHOW GRANTS FOR 'user1';
```

Result Grid | Filter Rows: | Export:

user
user1
departments_user
mysql.infoschema
mysql.session
mysql.sys
root

```
26    #t
27 •  SELECT user FROM mysql.user;
28 •  SHOW GRANTS FOR 'user1';
```

Result Grid | Filter Rows: | Export: | Wra

	Grants for user1@%
▶	GRANT USAGE ON *.* TO `user1`@`%`
	GRANT SELECT ON `lab_sheet_4`.`student` T...